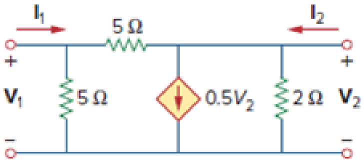
Chapter 19, Problem 22P

Problem

Obtain the *y* parameters of the two-port network in Fig. 19.83.

Figure 19.83



Step-by-step solution

Step 1 of 8

Consider the following circuit diagram:

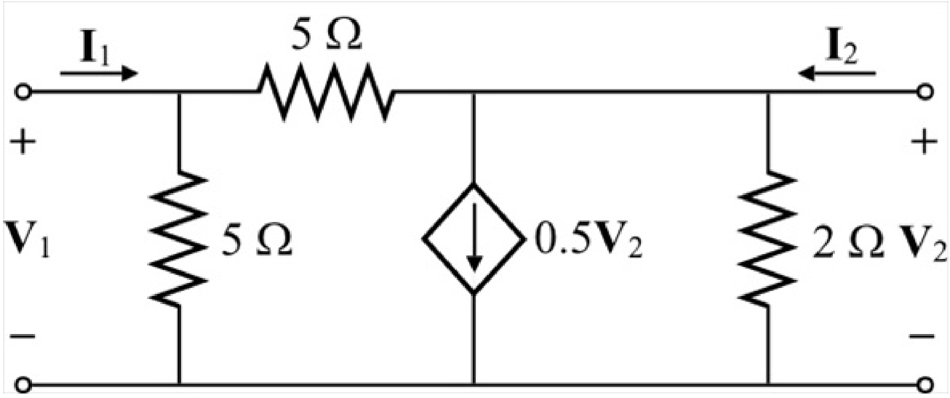


Figure 1

Step 2 of 8

Write the *y*-parameters.

$$\mathbf{y}_{11}=\left.\frac{\mathbf{I}_1}{\mathbf{V}_1}\right|_{\mathbf{V}_2=0} \quad \ldots\ldots (1)$$

$$\mathbf{y}_{12}=\left.\frac{\mathbf{I}_1}{\mathbf{V}_2}\right|_{\mathbf{V}_1=0} \quad \ldots\ldots (2)$$

$$\mathbf{y}_{21}=\left.\frac{\mathbf{I}_2}{\mathbf{V}_1}\right|_{\mathbf{V}_2=0} \quad \ldots\ldots (3)$$

$$\mathbf{y}_{22}=\left.\frac{\mathbf{I}_2}{\mathbf{V}_2}\right|_{\mathbf{V}_1=0} \quad \ldots\ldots (4)$$

Step 3 of 8

To calculate parameters $\mathbf{y}_{11}$ and $\mathbf{y}_{21}$, short circuit the output source, and connect a test voltage source to the input port. Draw the modified circuit diagram.

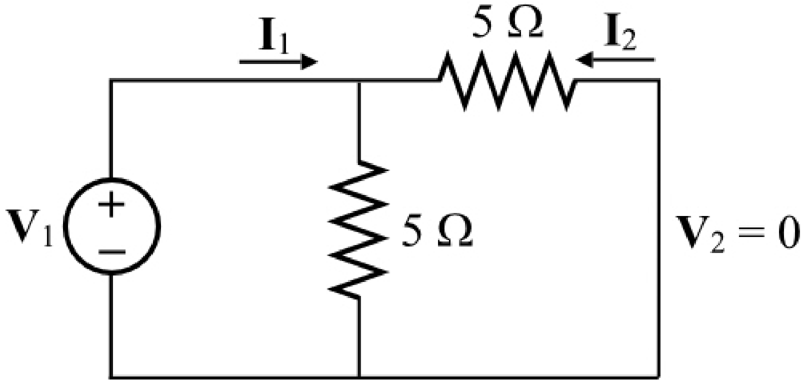


Figure 2

Step 4 of 8

Apply Kirchhoff's voltage law to the right side loop.

$$
\begin{aligned}
&5(\mathbf{I}_2-\mathbf{I}_1)+5\mathbf{I}_2=0\\
&-5\mathbf{I}_1+(5+5)\mathbf{I}_2=0\\
&10\mathbf{I}_2=5\mathbf{I}_1\\
&\mathbf{I}_2=\frac{\mathbf{I}_1}{2}
\end{aligned}
$$

Step 5 of 8

Apply Kirchhoff's voltage law to the left side loop.

$$
\begin{aligned}
\mathbf{V}_1&=5(\mathbf{I}_2+\mathbf{I}_1)\\
&=5\left(-\frac{\mathbf{I}_1}{2}+\mathbf{I}_1\right)\\
&=5\left(\frac{-\mathbf{I}_1+2\mathbf{I}_1}{2}\right)\\
&=\frac{5\mathbf{I}_1}{2}
\end{aligned}
$$

Step 6 of 8

Calculate the parameter $\mathbf{y}_{11}$.

$$\begin{aligned}\mathbf{y}_{11} &= \frac{\mathbf{I}_1}{\mathbf{V}_1} \\ &= \frac{\mathbf{I}_1}{\left(\frac{5\mathbf{I}_1}{2}\right)} \\ &= \frac{2}{5} \\ &= 0.4\ \text{S}\end{aligned}$$

Calculate the parameter $\mathbf{y}_{21}$.

$$\begin{aligned}\mathbf{y}_{21} &= \frac{\mathbf{I}_2}{\mathbf{V}_1} \\ &= \frac{\frac{\mathbf{I}_1}{2}}{\frac{5\mathbf{I}_1}{2}} \\ &= \frac{1}{5} \\ &= 0.2\ \text{S}\end{aligned}$$

To calculate the parameters $\mathbf{y}_{12}$ and $\mathbf{y}_{22}$, short the input port and connect a test voltage source to the output port.

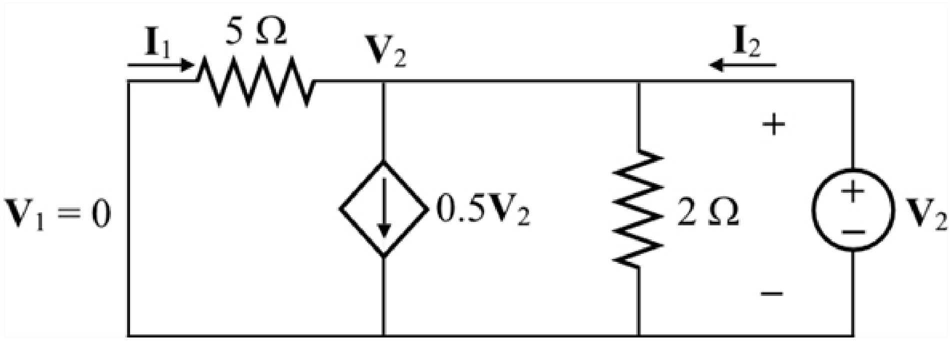


Figure 2

Step 7 of 8

Apply Kirchhoff's current law at node $\mathbf{V}_2$.

$$\frac{\mathbf{V}_2}{5} + 0.5\mathbf{V}_2 + \frac{\mathbf{V}_2}{2} = \mathbf{I}_2$$
$$(0.2 + 0.5 + 0.5)\mathbf{V}_2 = \mathbf{I}_2$$
$$1.2\mathbf{V}_2 = \mathbf{I}_2$$
$$\frac{\mathbf{I}_2}{\mathbf{V}_2} = 1.2\ \text{S}$$
$$\mathbf{y}_{22} = 1.2\ \text{S}$$

Step 8 of 8

Apply Kirchhoff's current law at node $\mathbf{V}_2$.

$$0.5\mathbf{V}_2 + \frac{\mathbf{V}_2}{2} = \mathbf{I}_1 + \mathbf{I}_2$$

$$0.5\mathbf{V}_2 + \frac{\mathbf{V}_2}{2} = \mathbf{I}_1 + 1.2\mathbf{V}_2$$

$$(0.5 + 0.5 - 1.2)\mathbf{V}_2 = \mathbf{I}_1$$

$$-0.2\mathbf{V}_2 = \mathbf{I}_1$$

$$\frac{\mathbf{I}_1}{\mathbf{V}_2} = -0.2\ \mathrm{S}$$

$$\mathbf{y}_{12} = -0.2\ \mathrm{S}$$

Therefore, the $\mathbf{y}$ parameters, $\begin{bmatrix} \mathbf{y}_{11} & \mathbf{y}_{12} \\ \mathbf{y}_{21} & \mathbf{y}_{22} \end{bmatrix}$ are $\boxed{\begin{bmatrix} 0.4 & -0.2 \\ 0.2 & 1.2 \end{bmatrix} \mathrm{S}}$.